

Chapter 4, Section 4

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3. Let the elliptic curve X be embedded in $\mathbf{P}^2$ so as to have the equation $y^2 = x(x-1)(x-\lambda)$. Show that any automorphism of X leaving $P_0 = (0 : 1 : 0)$ fixed is induced by an automorphism of $\mathbf{P}^2$ coming from the automorphism of the affine (x, y) -plane given by

$$\begin{cases} x' = ax + b \\ y' = cy. \end{cases}$$

In each of the four cases of (4.7), describe these automorphisms of $\mathbf{P}^2$ explicitly, and hence determine the structure of the group $G = \text{Aut}(X, P_0)$.

Proof. The embedding of X in $\mathbf{P}^2$ is given by the complete linear system $|3P_0| \cong \mathbf{P}^2$. Let $1, x, y$ be a k -basis of $H^0(X, \mathcal{O}_X(3P_0))$ such that x, y are local affine coordinates of $|3P_0|$. Any automorphism of X leaving P_0 induces an automorphism of $|3P_0|$, i.e., there exists a faithful representation $\rho : G \hookrightarrow \text{Aut}(|3P_0|) \cong \text{PGL}(2)$. If $\sigma \in G$, then we can write

$$\rho(\sigma) = \begin{pmatrix} a & 0 & b \\ c & d & e \\ f & 0 & g \end{pmatrix}, \quad d(ag - bf) \neq 0.$$

The unique g_2^1 , denoted $f : X \rightarrow \mathbf{P}^1$, is given by the projection of X onto the x -axis. By (4.4), there exists $\tau \in \text{PGL}(1)$ such that $\tau \circ f = f \circ \sigma$.

$j \neq 0, 1728$ The unique nontrivial element $\sigma \in G$ interchanges the sheets of the unique g_2^1 and can be written as

$$\begin{cases} x' = x \\ y' = -y \end{cases}.$$

$j = 1728$ and $\text{char } k \neq 3$

$j = 0$ and $\text{char } k \neq 3$

$j = 0$ ($= 1728$) and $\text{char } k = 3$

□

4. Let X be an elliptic curve in $\mathbf{P}^2$ given by an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

Show that the j -invariant is a rational function of the a_i , with coefficients in $\mathbf{Q}$. In particular, if the a_i are all in some field $k_0 \subseteq k$, then $j \in k_0$ also. Furthermore, for every $\alpha \in k_0$, there exists an elliptic curve defined over k_0 with j -invariant equal to α .

Proof. Complete the square in y to get an equation of the form

$$y^2 = x^3 + Ax^2 + Bx + C,$$

with

$$A = \frac{a_1^2}{4} + a_2, \quad B = \frac{a_1a_3}{2} + a_4, \quad C = \frac{a_3^2}{4} + a_6.$$

Let α, β, γ be the roots of the cubic on the right-hand side. Make a linear change of x to send α, β to 0, 1 so that the equation becomes

$$y^2 = x(x-1)(x-\lambda), \quad \lambda = \frac{\gamma - \alpha}{\beta - \alpha}.$$

Plugging in λ into the j -invariant formula gives

$$j = -2^8 \frac{(\alpha + \beta + \gamma)^2 - 3(\alpha\beta - \beta\gamma - \gamma\alpha)}{(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2}.$$

This is a rational function of the symmetric polynomials in α, β, γ , which are themselves polynomials in A, B, C , and hence in the a_i . $\square$

7. The Dual of a Morphism. Let X and X' be elliptic curves over k , with base points P_0, P'_0 .

- (a) If $f : X \rightarrow X'$ is any morphism, use (4.11) to show that $f^* : \text{Pic } X' \rightarrow \text{Pic } X$ induces a homomorphism $\hat{f} : (X', P'_0) \rightarrow (X, P_0)$. We call this the *dual* of f .
- (b) If $f : X \rightarrow X'$ and $g : X' \rightarrow X''$ are two morphisms, then $\widehat{g \circ f} = \hat{f} \circ \hat{g}$.
- (c) Assume $f(P_0) = P'_0$, and let $n = \deg f$. Show that if $Q \in X$ is any point, and $f(Q) = Q'$, then $\hat{f}(Q') = n_X(Q)$. Conclude that $f \circ \hat{f} = n_{X'}$ and $\hat{f} \circ f = n_X$.
- (d) If $f, g : X \rightarrow X'$ are two morphisms preserving the base points P_0, P'_0 , then $\widehat{f + g} = \hat{f} + \hat{g}$.
- (e) Using (d), show that for any $n \in \mathbf{Z}$, $\hat{n}_X = n_X$. Conclude that $\deg n_X = n^2$.
- (f) Show for any f that $\deg \hat{f} = \deg f$.

Proof.

- (a) Any morphism $f : X \rightarrow X'$ induces a natural transformation of functors $f^* : \text{Pic}^\circ(X'/-) \rightarrow \text{Pic}^\circ(X/-)$ (4.10.2). Under the identification $\text{Pic}^\circ(X/-) \cong \text{Hom}(-, J)$ and $\text{Pic}^\circ(X'/-) \cong \text{Hom}(-, J')$, Yoneda's lemma says there exists a natural bijection

$$\text{Hom}(J', J) \cong \text{Hom}(\text{Pic}^\circ(X'/-), \text{Pic}^\circ(X/-)),$$

which means there exists a unique $\hat{f} : J' \rightarrow J$ such that f^* is given by composition by $\hat{f}$. We obtain our desired morphism by identifying $X \cong J$ and $X' \cong J'$, and composing $\hat{f}$ by an automorphism of X that sends $f(P'_0)$ to P_0 (4.2).

- (b)
- (c)
- (d)
- (e) Apply (d) to $f = \text{id}_X$ and $\hat{1}_X = 1_X$ to get $\hat{n}_X = n_X$.
- (f)

$\square$

9. We say two elliptic curves X, X' are *isogenous* if there is a finite morphism $f : X \rightarrow X'$.

- (a) Show that isogeny is an equivalence relation.
- (b) For any elliptic curve X , show that the set of elliptic curves X' isogenous to X , up to isomorphism is countable.

Proof.

- (b)

$\square$

11. Let X be an elliptic curve over $\mathbf{C}$, defined by the elliptic functions with periods 1, τ . Let R be the ring of endomorphisms of X .

- (a) If $f \in R$ is a nonzero endomorphism corresponding to complex multiplication by α as in (4.18), show that $\deg f = |\alpha|^2$.
- (b) If $f \in R$ corresponds to $\alpha \in \mathbf{C}$ again, show that the dual $\hat{f}$ of (Ex. 4.7) corresponds to the complex conjugate $\bar{\alpha}$ of α .
- (c) If $\tau \in \mathbf{Q}(\sqrt{-d})$ happens to be integral over $\mathbf{Z}$, show that $R = \mathbf{Z}[\tau]$.

Proof.

- (a)
- (b)
- (c)

□

- 13.** If $p = 13$, there is just one value of j for which the Hasse invariant of the corresponding curve is 0. Find it.

Proof. The polynomial $h_p(\lambda)$ of (4.22) for $p = 13$ is

$$h_{13}(\lambda) = \lambda^6 + 10\lambda^5 + 4\lambda^4 + 10\lambda^3 + 4\lambda^2 + 10\lambda + 1,$$

and an elliptic curve has Hasse invariant 0 if and only if $h_{13}(\lambda) = 0$. On the otherhand, consider the j -invariant

$$j = 2^8 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}.$$

Expanding above modulo 13 gives

$$\lambda^6 + 10\lambda^5 + (10j + 6)\lambda^4 + (6j + 6)\lambda^3 + (10j + 6)\lambda^2 + 10\lambda + 1 = 0.$$

Matching coefficients with $h_{13}(\lambda)$ and solving for j gives $j \equiv 5 \pmod{13}$.

□

- 14.** The Fermat curve $X : x^3 + y^3 = z^3$ gives a nonsingular curve in characteristic p for every $p \neq 3$. Determine the set $\mathfrak{P} = \{p \neq 3 \mid X_{(p)} \text{ has Hasse invariant } 0\}$, and observe (modulo Dirichlet's theorem) that it is a set of primes of density $1/2$.

Proof. Using (4.21) with $f(x, y, z) = x^3 + y^3 - z^3$, we find that the coefficient of $(xyz)^{p-1}$ in f^{p-1} is zero if and only if $p \equiv 2 \pmod{3}$. By Dirichlet's theorem, the density of $\mathfrak{P}$ is $1/\phi(3) = 1/2$. □

- 15.** Let X be an elliptic curve over a field k of characteristic p . Let $F' : X_p \rightarrow X$ be the k -linear Frobenius morphism (2.4.1). Use (4.10.7) to show that the dual morphism $\hat{F}' : X \rightarrow X_p$ is separable if and only if the Hasse invariant of X is 1. Now use (Ex. 4.7) to show that if the Hasse invariant is 1, then the subgroup of points of order p on X is isomorphic to $\mathbf{Z}/p$; if the Hasse invariant is 0, then it is 0.

Proof.

□

- 16.** Again let X be an elliptic curve over k of characteristic p , and suppose X is defined over the field $\mathbf{F}_q$ of $q = p^r$ elements, i.e., $X \subseteq \mathbf{P}^2$ can be defined by an equation with coefficients in $\mathbf{F}_q$. Assume also that X has a rational point over $\mathbf{F}_q$. Let $F' : X_q \rightarrow X$ be the k -linear Frobenius with respect to q .

- (a) Show that $X_q \cong X$ as schemes over k , and that under this identification, $F' : X \rightarrow X$ is the map obtained by the q th-power map on the coordinates of points of X , embedded in $\mathbf{P}^2$.
- (b) Show that $1_X - F'$ is a separable morphism and its kernel is just the set $X(\mathbf{F}_q)$ of points of X with coordinates in $\mathbf{F}_q$.
- (c) Using (Ex. 4.7), show that $F' + \hat{F}' = a_X$ for some integer a , and that $N = q - a + 1$, where $N = \#X(\mathbf{F}_q)$.
- (d) Use the fact that $\deg(m + nF') > 0$ for all $m, n \in \mathbf{Z}$ to show that $|\alpha| \leq 2\sqrt{q}$. This is Hasse's proof of the analogue of the Riemann hypothesis for elliptic curves.

- (e) Now assume $q = p$, and show that the Hasse invariant of X is 0 if and only if $a \equiv 0 \pmod{p}$. Conclude that for $p \geq 5$ that X has Hasse invariant 0 if and only if $N = p + 1$.

Proof.

- (a)
- (b)
- (c)
- (d)
- (e)

□

17. Let X be the curve $y^2 + y = x^3 - x$ of (4.23.8).

- (a) If $Q = (a, b)$ is a point on the curve, compute the coordinates of the point $P + Q$, where $P = (0, 0)$, as a function of a, b . Use this formula to find the coordinates nP , $n = 1, 2, \dots, 10$.
- (b) This equation defines a nonsingular curve over $\mathbf{F}_q$ for all $p \neq 37$.

Proof.

- (a)
- (b)

□

20. Let X be an elliptic curve over a field k of characteristic $p > 0$, and let $R = \text{End}(X, P_0)$ be its ring of endomorphisms.

- (a) Let X_p be the curve over k defined by changing the k -structure of X (2.4.1). Show that $j(X_p) = j(X)^{1/p}$. Thus, $X \cong X_p$ over k if and only if $j \in \mathbf{F}_p$.
- (b) Show that p_X in R factors into a product of $\pi\hat{\pi}$ of two elements of degree p if and only if $X \cong X_p$. In this case, the Hasse invariant of X is 0 if and only if π and $\hat{\pi}$ are associates in R .
- (c) If $\text{Hasse}(X) = 0$ show in any case $j \in \mathbf{F}_{p^2}$.
- (d) For any $f \in R$, there is an induced map $f^* : H^1(\mathcal{O}_X) \rightarrow H^1(\mathcal{O}_X)$. This must be multiplication by an element $\lambda_f \in k$. So we obtain a ring homomorphism $\varphi : R \rightarrow k$ by sending f to λ_f . Show that any $f \in R$ commutes with the (nonlinear) Frobenius morphism $F : X \rightarrow X$, and conclude that if $\text{Hasse}(X) \neq 0$, then the image of φ is $\mathbf{F}_p$. Therefore, R contains a prime ideal $\mathfrak{p}$ with $R/\mathfrak{p} \cong \mathbf{F}_p$.

Proof.

- (a)
- (b)
- (c)
- (d)

□

22. If $X \rightarrow \mathbf{A}_C^1$ is a family of elliptic curves having a section, show that the family is trivial.

Proof.

□